

# TrES-1b

R-1683

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-1\_160822 · created 2026-08-02T12:05:02+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2457622.8267 +/- 0.0017 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.183 +/- 0.052                 |
| Transit depth (Rp/Rs)^2                           | 3.34 +/- 1.89 [%]               |
| Orbital Inclination (inc)                         | 89.97 +/- 1.41                  |
| Airmass coefficient 1 (a1)                        | 0.92 +/- 0.074                  |
| Airmass coefficient 2 (a2)                        | 0.027 +/- 0.011                 |
| Scatter in the residuals of the lightcurve fit is | 8.56 %                          |
| Best Comparison Star                              | #2 - [28, 210]                  |
| Optimal Aperture                                  | 5.68                            |
| Optimal Annulus                                   | 8.87                            |
| Transit Duration (day)                            | 0.101 +/- 0.0041                |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.183 ± 0.052 vs 0.1358 (deviation 0.91σ) |
| Duration fit vs published | 0.101 d vs 0.1042 d (deviation 0.77σ)     |
| Mid-time O–C              | -4.1 min                                  |
| Depth SNR                 | 3.5                                       |
| Residual scatter          | 8.56 %                                    |

## Light curve

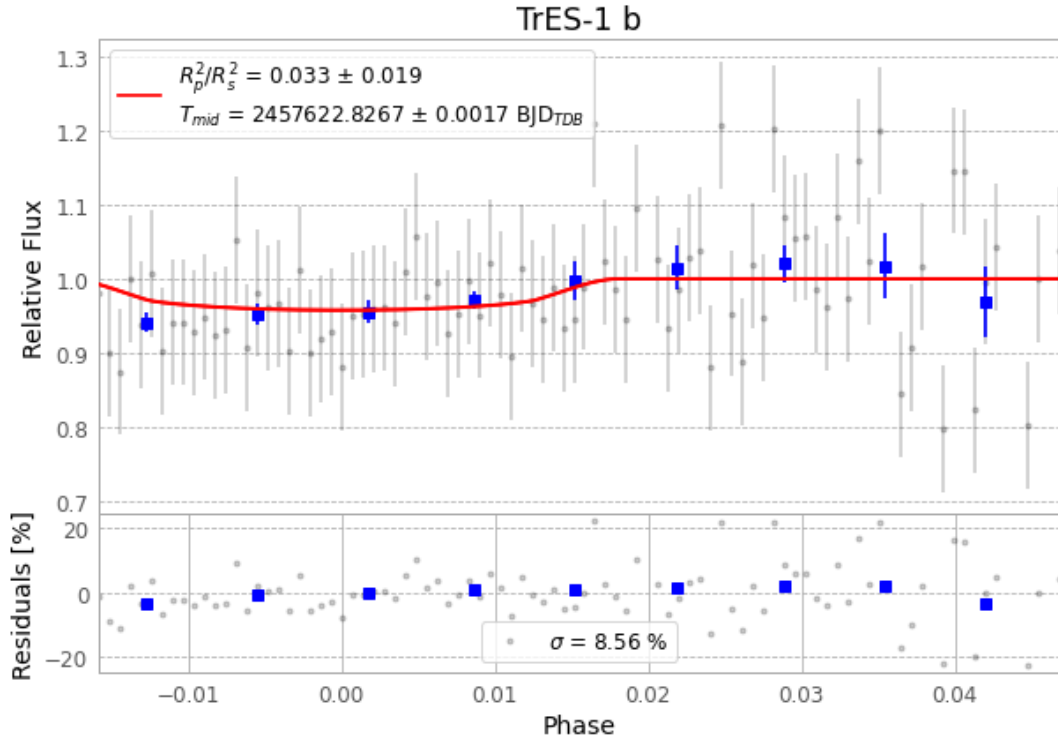


Figure 1 — FinalLightCurve\_TrES-1 b\_2016-08-21.png (SHA-256 7a327e8cfb1e40b2...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [38, 93], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 8e9878f1d0344633...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

92 FITS frames (60 MB), TRES-1160822063612.FITS → TRES-1160822110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-160822.log (80106c9e413d...), FinalLightCurve\_TrES-1 b\_2016-08-21.png (7a327e8cfb1e...), FinalLightCurve\_TrES-1 b\_2016-08-21.csv (2544841f5d50...)

## Compute resources

Wall time 345.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 12:05Z.